Certified Bounds for Partially Coherent Mask Synthesis: Convex Printability Certificates, Rigorously Separated Single- and Multi-Exposure Optima, and Pattern-Specific Exposure-Latitude Limits

Author names withheld (autonomous research system with human oversight)*

October 12, 2026 — manuscript v2 (internally verified; not peer reviewed)

Abstract

Purpose. Mask synthesis for projection lithography (ILT, SMO) is nonconvex, so a local optimum says nothing about the best achievable image. We provide solver-independent limits on printability.

Methods. For the Abbe/Hopkins model of a periodic cell—scalar or vector, with arbitrary aberrations and focus sets—the best achievable dark-to-bright intensity ratio $\rho^\star$ (equivalently, the maximum exposure latitude) is bounded by a semidefinite relaxation in the diffraction-order domain. Its dual is a certificate: nonnegative weights on dark and bright sample points such that one matrix is positive semidefinite, verified independently of the solver and rigorized against round-off. We prove that the relaxation equals the unlimited-multiple-exposure problem, that nonnegative one-dimensional masks form the Carathéodory–Toeplitz cone, and that pixelation at or above the collected-order count leaves the bound unchanged. An interval branch-and-bound certifier bounds the single-mask optimum.

Results. The bound reproduces the k_1 cutoffs for DUV and EUV. On 109 random one-dimensional targets, local optimization is certified globally optimal in 70% of cases and within 1% in 83%; all two-dimensional contact arrays are certified within 0.3%, and two of three 320 nm T-junction and line-end cells within 0.6%. Multi-exposure gains of 6.4–16.4% (random coherent targets) and 13.4% (mixed-width target) are rigorously certified. The bound is 3–150× tighter than the Rayleigh-quotient bound, certifies that binary masks forfeit 3–14% relative to phase-shift masks, and shows that a standard pattern-error ILT lands 30–67% above the certified optimum.

Conclusions. The certificates make local optimality checkable, quantify multi-exposure and phase-shift entitlement, and survive the vector model and moderate aberrations.

1 Introduction

Computational lithography synthesizes a mask (and a source) so that the projected aerial image prints a target layout. In the Abbe formulation the image is a weighted sum, over source points, of squared moduli of filtered mask spectra: it is *quadratic* in the mask. ILT and SMO therefore optimize nonconvex functionals. IBM's wavefront-optimization work states that mask optimization "tends inherently to be a problem with non-convex quadratic constraints" [1]; recent work on learned ILT describes objectives "that often trap optimization in poor local minima" [2]; and reviews list local optimality among the open problems [3, 4]. What is missing is a *certificate*: a proof, for a given target, illumination, numerical aperture and wavelength, that

*Code, data and the full verification record are available in the accompanying repository (`code/`, `results/`, `notes/`). Every number in this manuscript is produced by `code/run_all_par.sh` from scratch.

no mask can do better than a stated value. Such certificates (i) separate unprintable layouts from unconverged optimizers, (ii) benchmark ILT engines against the true optimum, (iii) quantify the value of multiple exposure and of phase-shift versus binary masks, (iv) generalize the k_1 rule to arbitrary layouts, and (v) localize the geometry that limits printability through the support of the dual weights.

Contributions.

(C1) Theorem 1: dual certificates of printability, valid for any mask class and for incoherent sums of masks, verified by one PSD test independent of the solver, together with a rigorization procedure (§5).

(C2) Theorem 3: the relaxation equals the unlimited-multi-exposure problem, and Pataki's bound limits the number of useful exposures. Lemma 6: complex transmission cannot improve the relaxation under symmetric illumination. Theorem 4: Carathéodory–Toeplitz realizability for nonnegative one-dimensional masks. Lemmas 7–8: vector imaging, aberrations and pixelation preserve the structure.

(C3) Theorem 9: a rigorous branch-and-bound certificate of the single-mask optimum which, combined with (C1), proves strictly positive multi-exposure gains on concrete targets.

(C4) An open pipeline and an empirical study with independent verification of every certificate.

Not claimed. No new SDP algorithm (the Shor relaxation of a QCQP is textbook [5, 6]); no full-chip scalability; no thick-mask electromagnetics or resist chemistry; no statement about proprietary ILT engines.

2 Related Work

Wavefront optimization. Rosenbluth et al. [1, 7] optimize the collected diffraction orders directly, maximizing exposure latitude subject to bright/dark intensity constraints; a patent of the same group writes the problem as a quadratic program in the order amplitudes [8]. Their global strategies partition the solution space into regions with at most one local minimum [1], use dominant "joint eigenvectors" and a "step-walk relaxation", and, for multiple exposure, an unconstrained Rayleigh-quotient simplification [8]. None yields a bound on solutions obtained by other means; we compare with the Rayleigh-quotient bound in Table 8.

Exact methods for small binary masks. Liu and Zakhor [9] designed binary and phase-shift masks by branch and bound on a pixel least-squares criterion (combinatorial), whereas Theorem 9 certifies the continuous single-mask optimum of the ratio problem.

SDP relaxation of QCQPs. The Shor relaxation and its approximation theory are surveyed in [5]; max-min multicast beamforming [10] has exactly the structure of coherent-illumination mask synthesis. Pataki [11] bounds the rank of extreme solutions; Lasserre's hierarchy [12] tightens bounds for small problems.

Bounds in photonic design. Angeris, Vučković and Boyd [13, 14] bound inverse-designed photonic devices by Lagrangian duality; our work is the imaging-side analogue. SDP relaxations have been used in lithography only for the combinatorial mask-assignment problem of multiple patterning [15].

Information-theoretic limits and lifting. Information-theoretic limits [16, 17] bound OPC fidelity through mutual information; these are not layout-specific duality certificates. PhaseLift [18] lifts quadratic measurements for recovery; we lift for synthesis. The optimal coherent decomposition of partially coherent imaging is due to [19].

3 Model and Problem Statement

3.1 Imaging model

Let $\Lambda = [0, L_1) \times \cdots \times [0, L_d)$, $d \in \{1, 2\}$, be a periodic cell and $m : \Lambda \to \mathbb{C}$ the mask transmission with Fourier coefficients $M_k = |\Lambda|^{-1} \int_\Lambda m(x) e^{-2\pi i f_k \cdot x}\, dx$, $f_k = (k_1/L_1, \dots)$. Illumination is a finite set of source points $s_j \in \mathbb{R}^d$ (direction cosines, $|s_j| \le \sigma\,\mathrm{NA}$) with weights $w_j > 0$, $\sum_j w_j = 1$. The pupil is $P_z(g) = \mathbb{1}\{|g|\lambda \le \mathrm{NA}\} \exp\left(i2\pi z(\sqrt{1 - |g|^2\lambda^2} - 1)/\lambda + iW(g)\right)$ with defocus z and aberration phase W. Only the collected orders $K = \{k : |f_k| \le (\mathrm{NA} + \max_j |s_j|)/\lambda\}$, $F = |K|$, contribute; $M \in \mathbb{C}^F$ denotes $(M_k)_{k \in K}$. The Abbe image at x and focus z is

$$I_{M,z}(x) = \sum_j w_j \Big| \sum_{k \in K} P_z(f_k + s_j/\lambda)\, M_k\, e^{2\pi i f_k \cdot x} \Big|^2 = M^{\mathsf{H}} Q_{x,z} M, \qquad Q_{x,z} = \sum_j w_j c_{j,x,z} c_{j,x,z}^{\mathsf{H}} \succeq 0. \tag{1}$$

3.2 Synthesis problem and relaxation

A target is specified by finite sample sets B (bright) and D (dark), placed at exactly $\pm\tau$ (edge tolerance) from every target edge plus interior points, and by a focus set Z. Mask classes: $\mathcal{M}_{\mathbb{C}} = \mathbb{C}^F$; real masks $\mathcal{M}_{\mathbb{R}} = \{M : M_{-k} = \overline{M_k}\}$ (phase-shift masks); nonnegative masks $\mathcal{M}_+ \subset \mathcal{M}_{\mathbb{R}}$ (binary/attenuated). The dark-to-bright ratio

$$\rho(M) = \frac{\max_{(x,z) \in D \times Z} I_{M,z}(x)}{\min_{(x,z) \in B \times Z} I_{M,z}(x)} \tag{2}$$

is scale-invariant; for a threshold resist with one dose over the focus set, $\mathrm{EL}(\rho) = (1 - \rho)/(1 + \rho)$ is the exposure latitude. The synthesis problem and its relaxation are

$$\begin{aligned} (\mathrm{P}_{\mathcal{M}}) \quad \rho^\star(\mathcal{M}) &= \inf_{M \in \mathcal{M} \setminus \{0\}} \rho(M) \\ &= \min\left\{t : \ M^{\mathsf{H}} Q_d M \le t \ \ \forall d \in D \times Z, \ \ M^{\mathsf{H}} Q_b M \ge 1 \ \ \forall b \in B \times Z\right\}, \qquad (3) \\ (\mathrm{R}) \quad t_R &= \min\left\{t : \ \mathrm{Tr}(Q_d X) \le t \ \ \forall d \in D \times Z, \ \ \mathrm{Tr}(Q_b X) \ge 1 \ \ \forall b \in B \times Z, \right. \\ &\qquad \left. X \succeq 0 \ \ [\, X = \mathrm{conj}(\Pi X \Pi)\,]\right\}, \qquad (4) \end{aligned}$$

where Π is the order-flip permutation and the bracketed constraint selects real masks.

3.3 Notation

Throughout, ρ is the dark-to-bright intensity ratio of a given mask, $\rho^\star$ its infimum over a mask class, and t_R the value of the relaxation (R); "bound" always denotes the rigorized certificate value (a lower bound on $\rho^\star$, hence an upper bound on exposure latitude), and "best single mask" the best feasible real mask found by local search. The relative *gap* is (best single mask − bound)/bound. Subscripted λ_b denotes a dual weight on a bright sample (as in the figure legends), whereas unsubscripted λ is the wavelength. Table 1 collects the remaining symbols.

Table 1: Notation. Intensities are in arbitrary units (the ratio ρ is dimensionless); lengths in nm unless stated.

Symbol	Meaning
Λ, L_i, d	periodic cell, its side lengths, and its dimension ($d \in \{1, 2\}$)
m, M_k, M	mask transmission, its Fourier coefficients, and the vector of collected coefficients
K, $F = \lvert K\rvert$	set and number of collected diffraction orders
λ, NA, σ	wavelength, numerical aperture, partial-coherence factor
s_j, w_j	source points (direction cosines) and their weights
P_z, z, W	pupil function, defocus, aberration phase
Z	finite set of focus conditions
$I_{M,z}(x)$, $Q_{x,z}$	aerial image at (x, z) and its Hermitian PSD form in the orders
B, D, τ	bright and dark sample sets; edge tolerance (offset of samples from target edges)
$\mathcal{M}_{\mathbb{C}}$, $\mathcal{M}_{\mathbb{R}}$, $\mathcal{M}_{+}$	complex, real (phase-shift) and nonnegative (binary/attenuated) mask classes
ρ, $\rho^\star$, t_R	dark-to-bright ratio, its optimum over a mask class, value of the relaxation (R)
EL	exposure latitude $(1-\rho)/(1+\rho)$ of a threshold resist
μ_d, λ_b	dual (certificate) weights on dark and bright samples
S, $\tilde{S}$	certificate matrix $\sum_d \mu_d Q_d - \sum_b \lambda_b Q_b$ and its real-structured form
Π, J, U	order-flip permutation, antilinear involution $\mathrm{conj}(\Pi\,\cdot\,)$, unitary basis of $\mathcal{M}_{\mathbb{R}}$
X	lifted (PSD) matrix variable of the relaxation
κ	rigorization scale of a certificate (§5)
L, U (Theorem 9)	certified single-mask lower bound and incumbent upper bound
k_1	normalized half-pitch, $k_1 = \text{half-pitch} \cdot \mathrm{NA}/\lambda$

4 Theory

Theorem 1 (Certificate)**.** *Let $\mu \geq 0$ on $D \times Z$ with $\sum \mu = 1$, $\lambda \geq 0$ on $B \times Z$, and $S(\mu, \lambda) = \sum_d \mu_d Q_d - \sum_b \lambda_b Q_b \succeq 0$. Then $\rho(M) \geq \sum_b \lambda_b$ for every $M \in \mathbb{C}^F \setminus \{0\}$ and for every incoherent sum of images $\sum_i I_{M_i}$. For real masks it suffices that $\tilde{S} = \mathrm{Re}(U^{\mathsf{H}} S U) \succeq 0$ with U a unitary basis of $\mathcal{M}_{\mathbb{R}}$.*

Proof. Let $\beta = \min_{B\times Z} I$ and $\delta = \max_{D\times Z} I$. Then $\delta = \delta \sum_d \mu_d \geq \sum_d \mu_d M^{\mathsf{H}} Q_d M = M^{\mathsf{H}} S M + \sum_b \lambda_b M^{\mathsf{H}} Q_b M \geq \beta \sum_b \lambda_b$. For sums of images apply the same chain termwise. For $M = Ur$, $r \in \mathbb{R}^F$, $M^{\mathsf{H}} S M = r^{\mathsf{T}} \tilde{S} r$. □

Lemma 2 (Real structure)**.** *$J(M) = \mathrm{conj}(\Pi M)$ is an antilinear involution; $\mathcal{M}_{\mathbb{R}} = \mathrm{Fix}(J)$ is a real F-dimensional subspace on which the Hermitian form is real. A PSD X satisfies $X = \mathrm{conj}(\Pi X \Pi)$ iff $XJ = JX$; then $X = UYU^{\mathsf{H}}$ with $Y \in \mathbb{S}_+^F$ and $\mathrm{Tr}(QX) = \mathrm{Tr}(\tilde{Q}Y)$, $\tilde{Q} = \mathrm{Re}(U^{\mathsf{H}} Q U)$. Hence (R) for real masks is a real SDP of the same size.*

Theorem 3 (Relaxation equals unlimited multiple exposure)**.** *For $\mathcal{M} \in \{\mathcal{M}_{\mathbb{C}}, \mathcal{M}_{\mathbb{R}}\}$, t_R equals the optimal ratio of the summed image of any number of incoherently superposed exposures of masks in $\mathcal{M}$. If (R) has a rank-one optimum then $\rho^\star = t_R$ is attained by a single mask; conversely, if $\rho^\star = t_R$ is attained then (R) has a rank-one optimum. An extreme optimal solution has $\mathrm{rank}\, X = r$ with $r(r+1)/2 \leq |D \times Z| + |B \times Z|$, so at most that many exposures are needed.*

Proof. $X \succeq 0 \iff X = \sum_{i \leq F} M_i M_i^{\mathsf{H}}$, and $\mathrm{Tr}(QX) = \sum_i M_i^{\mathsf{H}} Q M_i$ is the summed image; Lemma 2 gives real M_i in the real-structured case. Writing the inequalities with slacks, (R) is a standard-form SDP with $m = |D \times Z| + |B \times Z|$ equalities and block-diagonal cone $(X, \text{slacks}, t^+, t^-)$; Pataki's theorem [11] bounds $\sum_{\text{blocks}} r_b(r_b+1)/2 \leq m$. □

Remark. Attainment of $\rho^\star$ is not automatic: ρ is scale-invariant and continuous on the unit sphere away from $\{\min_B I = 0\}$, and the infimum may be approached by masks whose bright and dark intensities vanish together; $\rho^\star$ is nevertheless certified by Theorem 1.

Theorem 4 (Nonnegative masks in one dimension)**.** *For $d = 1$ and $K = \{-K_{\max}, \dots, K_{\max}\}$, M is the in-band coefficient vector of a nonnegative measure iff the Toeplitz matrix $T(M) = (M_{a-b})_{a,b=0}^{K_{\max}} \succeq 0$; if $T(M) \succ 0$ it is realized by a bounded, strictly positive function. Hence $\overline{\mathcal{M}_+} = \{M : T(M) \succeq 0\}$.*

Proof. Necessity: $v^{\mathsf{H}} T(M) v = |\Lambda|^{-1} \int m(x) |\sum_a v_a e^{2\pi i a x/L}|^2 dx \geq 0$. Sufficiency is the Carathéodory–Toeplitz theorem [20]; measures are weak-$*$ limits of bounded nonnegative functions (Fejér means). For $T(M) \succ 0$ the Levinson–Durbin recursion gives the maximum-entropy autoregressive extension $f(\theta) = \sigma^2/|a(e^{i\theta})|^2$ (here σ^2 is the innovation variance, unrelated to the partial-coherence factor) with a zero-free on the closed disk, matching $M_0, \dots, M_{K_{\max}}$; f is continuous, bounded and bounded away from zero. □

Proposition 5 (Cutoff)**.** *If $L(\mathrm{NA} + \max_j |s_j|)/\lambda < 1$ (pitch below $\lambda/((1+\sigma)\mathrm{NA})$) then $Q_{x,z}$ is independent of x, $t_R = 1$, and uniform (μ, λ) is a certificate. Above this pitch $t_R < 1$ for any target with a bright and a dark point.*

Lemma 6 (Complex versus real masks)**.** *If $w(-s) = w(s)$ and the pupil is even, then $\tilde{M} = \mathrm{conj}(\Pi M)$ produces the same image as M for all x, z; consequently $t_R(\mathcal{M}_{\mathbb{C}}) = t_R(\mathcal{M}_{\mathbb{R}})$ (take $X = \frac{1}{2}(MM^{\mathsf{H}} + \tilde{M}\tilde{M}^{\mathsf{H}})$).*

Lemma 7 (Vector imaging and aberrations)**.** *With the image-side plane wave of direction cosines (α, β, γ) carrying the mask-side polarization p rotated by the minimal rotation taking $\hat{z}$ to the propagation direction (for $p = \hat{x}$: $E = (1-\alpha^2/(1+\gamma),\ -\alpha\beta/(1+\gamma),\ -\alpha)$, Rodrigues' formula [21]), the image is a sum of rank-one Hermitian forms over channels (source, polarization, field component), so Theorems 1–3 apply verbatim; Lemma 6 extends to reflection-symmetric polarization ensembles since $E_{x,y}(-g) = E_{x,y}(g)$, $E_z(-g) = -E_z(g)$.*

Lemma 8 (Pixelation)**.** *For a mask piecewise constant on $N \geq F$ pixels per period (one dimension), $M_k = \mathrm{sinc}(k/N)\,\mathrm{DFT}_k(m)$ maps onto $\mathbb{C}^F$, so the pixel-domain and Fourier-domain relaxations coincide for the scale-free ratio problem; $N < F$ can only raise the bound.*

Remark (Sampled versus continuous constraints). The continuous problem has more constraints than the sampled one, so a certificate for the sampled problem is a valid lower bound for the continuous problem. Numerically, the interior samples carry zero dual weight and the bound is invariant to the interior spacing (§6.9).

Theorem 9 (Rigorous single-mask lower bound)**.** *Let $Q_1, \dots, Q_m \succeq 0$ (dark) and $P_1, \dots, P_n \succeq 0$ (bright) be real symmetric and $\rho(r) = \max_i q_i(r)/\min_j p_j(r)$. (i) $\rho^\star(\mathcal{M}_{\mathbb{R}}) = \min$ over the faces $\Phi_i = \{r : r_i = 1, |r_j| \leq 1\}$. (ii) For a box $\{c + \delta : |\delta_j| \leq h_j\}$ and $q(r) = r^{\mathsf{T}} Q r$ with $Q \succeq 0$: $q(c) - 2|Qc| \cdot h \leq q(r) \leq q(c) + 2|Qc| \cdot h + h^{\mathsf{T}}|Q|h$. (iii) Hence $\inf_{box} \rho \geq \mathrm{LB} := \max_i \max(q_i(c) - 2|Q_i c| \cdot h, 0)/\min_j (p_j(c) + 2|P_j c| \cdot h + h^{\mathsf{T}}|P_j|h)$, and a branch-and-bound over boxes covering $\bigcup_i \Phi_i$ returns, at any stopping time, a valid lower bound $L = \min_{leaves} \mathrm{LB} \leq \rho^\star(\mathcal{M}_{\mathbb{R}})$; on convergence $L \geq (1 - \mathrm{tol})\, U$ with U the incumbent.*

Proof. (i) Scale and sign invariance. (ii) $q(c+\delta) = q(c) + 2(Qc)\cdot\delta + \delta^{\mathsf{T}} Q \delta$ with $0 \leq \delta^{\mathsf{T}} Q \delta \leq h^{\mathsf{T}}|Q|h$. (iii) Interval arithmetic and the covering argument. □

5 Algorithms

Bound. Build $Q_{x,z}$ on the edge-offset sample set and solve the real SDP (CLARABEL [22]; SCS [23] as fallback).

Rigorization. Extract the duals and let $\mathcal{N}$ be the common null space of $S_D = \sum_d \mu_d Q_d$ and $S_B = \sum_b \lambda_b Q_b$ (orders invisible to every weighted constraint; on $\mathcal{N}$ every form in Theorem 1 vanishes identically). Find by bisection the largest $\kappa \le 1$ with $\lambda_{\min}(S_D - \kappa S_B) \ge 10^{-10}\|S_D - \kappa S_B\|$ on $\mathcal{N}^{\perp}$ (if S_B does not vanish on the near-null space of S_D, no projection is applied and the bisection lowers κ, to 0 if needed), and report $\kappa \sum_b \lambda_b$. Independent check: a 50-digit Cholesky factorization of the projected matrix (mpmath).

Primal. Gaussian rounding from X; analytic-gradient L-BFGS on a log-sum-exp surrogate of $\log \rho$ (common dose across foci); vectorized SLSQP minimax polish; multistart; optimal-face exploration (random PSD objectives over the optimal face) for low-rank optima; explicit two-exposure refinement.

Single-mask certification. Theorem 9 (batched boxes; error terms inflated by $1 + 10^{-9}$, 10^{-12} subtracted) and, as corroboration, the Lasserre-2 moment relaxation [12].

Variants. SMO relaxation (one PSD block per source point: unlimited exposures with coherent point sources); doubly-nonnegative (DNN) relaxation for nonnegative masks in the pixel domain; complex-mask relaxation; multi-focus; vector and aberrated pupils; pixel-integrated masks.

Complexity. $F \approx (2(1+\sigma)\mathrm{NA}L/\lambda)^d$; the SDP has an $F \times F$ matrix variable and $|B| + |D|$ constraints per focus. Solve times are 0.1–1.5 s for $F \le 15$ and 100–650 s for $F = 61$ with 1,200 constraints.

6 Experiments

Settings. DUV: $\lambda = 193\,\mathrm{nm}$, NA $= 1.35$, with coherent illumination; conventional $\sigma \in \{0.5, 0.7, 0.9\}$; dipole $\sigma \in \{0.7, 0.8\}$; annular 0.6/0.9; quasar 0.6/0.9. EUV: $\lambda = 13.5\,\mathrm{nm}$, NA $= 0.33$. Edge tolerance $\tau = 8\,\mathrm{nm}$ unless stated. "Bound" is always the rigorized certificate value and "best single mask" the best feasible real mask found (§3.3). All tables are generated from the clean reproduction (`results/RESULTS_TABLES.md`).

6.1 Resolution cutoff

For dense lines the bound is exactly 1 for pitches below $\lambda/((1+\sigma)\mathrm{NA}) = 84.1\,\mathrm{nm}$ at $\sigma = 0.7$ and drops below 1 immediately above (Fig. 1). In EUV with $\pm 50\,\mathrm{nm}$ focus, the 20 nm pitch ($k_1 = 0.244$) is certified unprintable with any mask under both sources; the $\sigma = 0.8$ dipole is also certified unprintable at 22 nm pitch, whereas the $\sigma = 0.9$ conventional source reaches EL $= 0.36$. All fourteen EUV cases are tight (gap 0.00%).

6.2 One-dimensional feature library

Dipole and dense-line cases are exactly tight; conventional-illumination gaps are at most 2.5%; coherent illumination shows large gaps (iso-dense 30.8%, mixed 13.7%), and the three-focus cases show gaps of 11–16% (Table 2). Figure 2 shows a certificate for the iso-dense target: the dual weights concentrate on the edge-offset samples, and the interior samples carry zero weight.

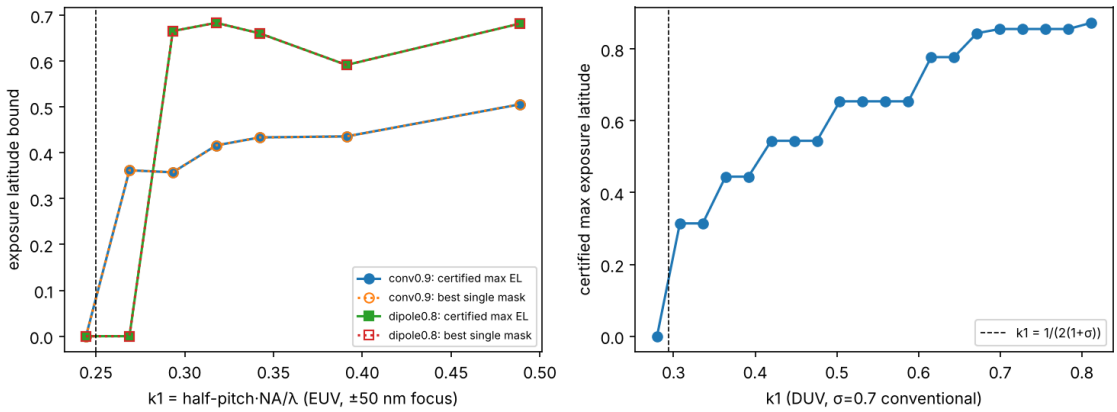


Figure 1: Certified maximum exposure latitude versus the normalized half-pitch $k_1 = \text{half-pitch} \cdot \text{NA}/\lambda$ for dense lines. Left: EUV ($\lambda = 13.5\,\text{nm}$, NA $= 0.33$) with $\pm 50\,\text{nm}$ focus under conventional $\sigma = 0.9$ and dipole $\sigma = 0.8$ illumination; the certified bound and the best single mask coincide on every point, and the dashed line marks $k_1 = 0.25$. Right: DUV ($\lambda = 193\,\text{nm}$, NA $= 1.35$) with conventional $\sigma = 0.7$ illumination; the dashed line is the classical cutoff $k_1 = 1/(2(1+\sigma))$, below which the certified exposure latitude is exactly zero (Proposition 5).

Table 2: One-dimensional feature library (DUV, $\lambda = 193\,\text{nm}$, NA $= 1.35$; edge tolerance $\tau = 8\,\text{nm}$, except 6 nm for the dense-90 case). F: number of collected orders; foci: $|Z|$, with $\pm 80\,\text{nm}$ denoting the focus set $\{-80, 0, +80\}$ nm; bound ρ: rigorized certificate value (dimensionless dark-to-bright ratio); max EL: $(1-\rho)/(1+\rho)$ at the bound; best single ρ: best feasible real mask found; gap = (best single mask − bound)/bound. "iso-dense" and "mixed" are the isolated-plus-dense and mixed-width line targets; "dense 90" is a 90 nm-pitch dense-line target, listed for conventional $\sigma = 0.7$ / dipole / coherent illumination.

Case	F	Foci	Bound ρ	Max EL	Best single ρ	Gap
iso-dense, coherent	9	1	0.3625	0.468	0.4741	30.8%
mixed, coherent	7	1	0.4211	0.407	0.4787	13.7%
iso-dense, $\sigma = 0.7$	15	1	0.4691	0.361	0.4808	2.5%
mixed, $\sigma = 0.7$	11	1	0.4825	0.349	0.4836	0.2%
iso-dense, dipole 0.8	15	1	0.4313	0.397	0.4313	0.0%
mixed, dipole 0.8	13	1	0.4211	0.407	0.4211	0.0%
iso-dense, $\sigma = 0.7$, ± 80 nm	15	3	0.8131	0.103	0.9044	11.2%
mixed, $\sigma = 0.7$, ± 80 nm	11	3	0.8691	0.070	0.9742	12.1%
iso-dense, dipole 0.8, ± 80 nm	15	3	0.7686	0.131	0.8921	16.1%
dense 90, $\sigma = 0.7$ / dipole / coherent	9/9/5	1	0.6428/0.4217/0.4217	0.217/0.407/0.407	= bound	0.0%

6.3 Random one-dimensional targets

On 109 random targets (1–4 segments of width $\geq 40\,\text{nm}$ in 300–600 nm cells; $\tau = 10\,\text{nm}$) the best single mask matches the certified bound to 10^{-4} in 69.7% of instances, to 1% in 82.6% and to 5% in 91.7% (median gap 1.8×10^{-6}, maximum 17.3%; Table 3, Fig. 3). Exact tightness is source-dependent (85% for $\sigma = 0.9$, 75% for the dipole, 57% for coherent and for $\sigma = 0.5$ illumination); every gap above 5% has minimal optimal-face rank 2 or 3, consistent with Theorem 3.

6.4 Certified multi-exposure gains

Theorem 9 converges in 0.3–1.6 s for $F = 5$ and proves the gains of Table 4 (Fig. 4, left); each optimal face has rank 2, so two exposures suffice. The Lasserre-2 single-mask bounds coincide with U to 10^{-6} on these instances (cross-checked between CLARABEL, 0.4787045, and SCS, 0.4787037, on the mixed target). For $F = 9$ (iso-dense, coherent) and $F = 11$ (three-focus mixed target) the interval branch-and-bound does not converge within 6×10^6 nodes; only Lasserre-2 values (0.4608 and 0.9552, solver status *optimal_inaccurate*) corroborate gains of 27% and 9.9%, which we therefore do not count as certified.

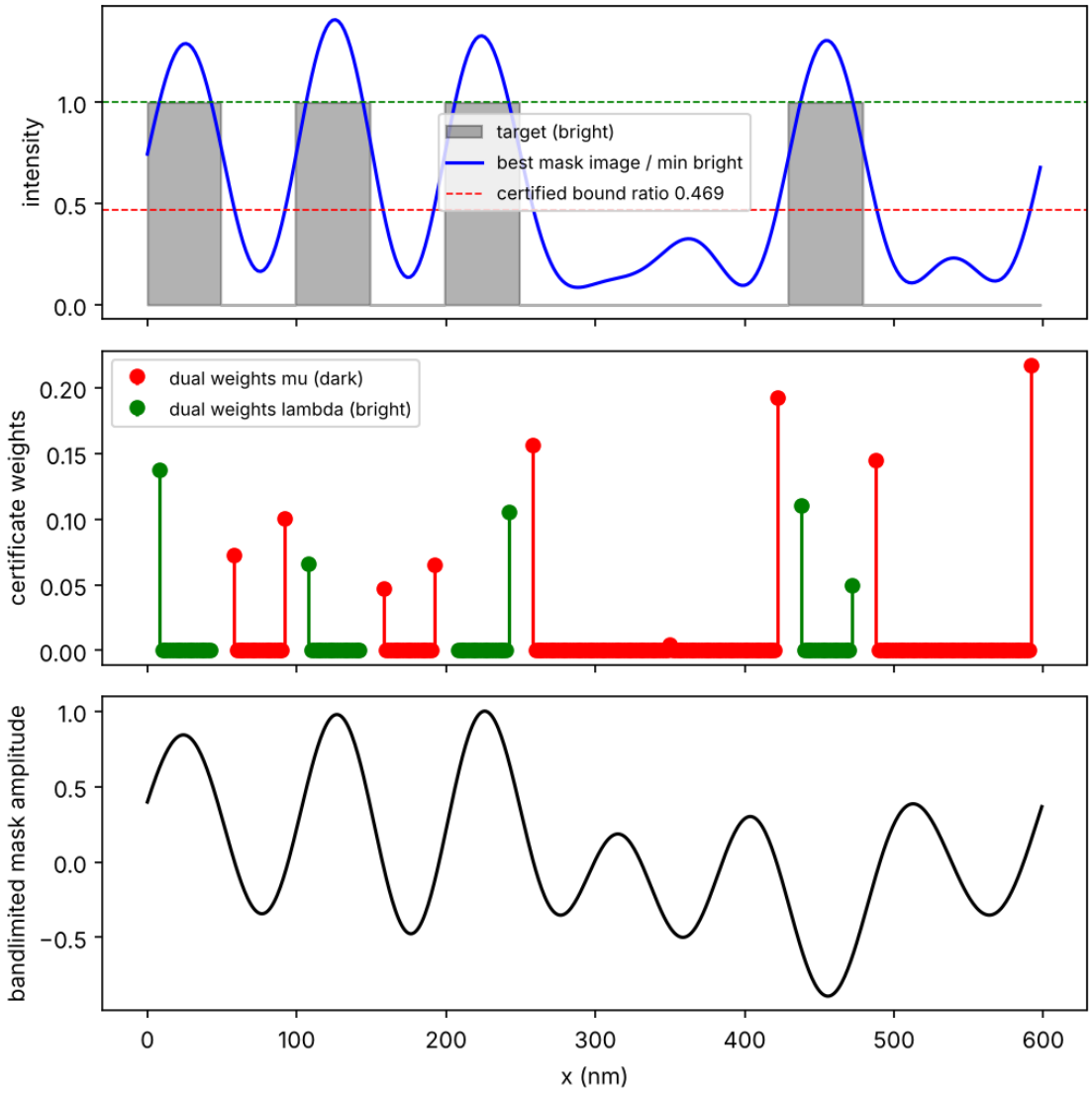


Figure 2: Certificate for the iso-dense target under conventional $\sigma = 0.7$ illumination (DUV, NA $= 1.35$; Table 2). Top: target (gray, bright regions) and the image of the best single mask normalized to its minimum bright intensity; the dashed red line is the certified bound $\rho = 0.469$, which no mask can beat. Middle: dual certificate weights μ_d (dark, red) and λ_b (bright, green); the weights concentrate on the edge-offset sample points and vanish on interior samples. Bottom: the band-limited amplitude of the best mask (a phase-shift mask; negative values are π-phase regions).

6.5 Phase-shift versus binary masks and pattern-error ILT

Table 5 compares the certified optima of phase-shift (PSM) and binary masks with the results of a standard pattern-error ILT on the iso-dense, mixed and dense-120 targets ($\sigma = 0.7$, $\tau = 10\,\text{nm}$). The DNN relaxation for nonnegative masks gives 0.4266/0.4599/0.3777 against PSM bounds of 0.3879/0.4031/0.3671: binary masks forfeit a certified 10%/14%/3%. Pixel- and Fourier-domain PSM relaxations agree to 10^{-5}. A standard pattern-error ILT (sigmoid resist, open-frame-normalized intensity, threshold 0.3, best of six settings, 1,500 Adam steps, gradient verified by finite differences) reaches 0.553/0.672/0.477 (PSM) and 0.579/0.722/0.517 (binary), i.e. 30–67% (PSM) and up to 79% (binary) above the certified PSM optimum. Refining the ILT mask on the ratio objective closes the gap on dense-120 (0.3671) and partially elsewhere (0.426 and 0.530).

Table 3: Tightness of the certified bound on 109 random one-dimensional targets (DUV, NA = 1.35; 1–4 segments of width $\geq 40\,\text{nm}$ in 300–600 nm cells; $\tau = 10\,\text{nm}$). Gap = (best single mask − bound)/bound. "Exactly tight" means gap $\leq 10^{-4}$. The per-source rows give the fraction of exactly tight instances for each illumination.

Criterion	Fraction of instances
gap $\leq 10^{-4}$ (exactly tight)	69.7%
gap $\leq 1\%$	82.6%
gap $\leq 5\%$	91.7%
median gap / maximum gap	1.8×10^{-6} / 17.3%
exactly tight, conventional $\sigma = 0.9$	85%
exactly tight, dipole	75%
exactly tight, coherent	57%
exactly tight, conventional $\sigma = 0.5$	57%

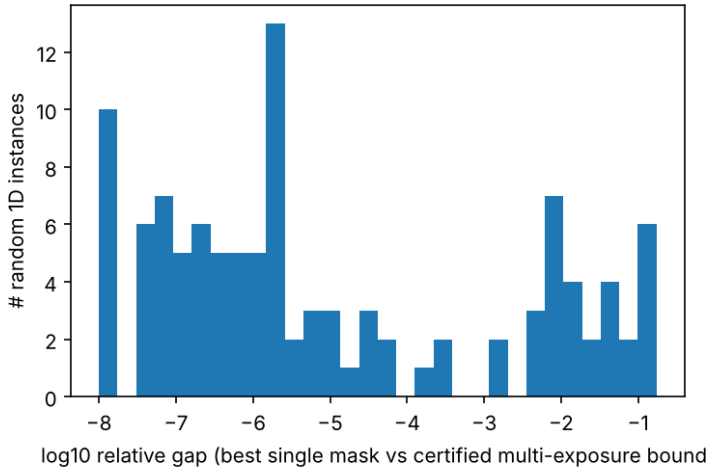


Figure 3: Histogram of $\log_{10}$ of the relative gap between the best single mask and the certified multi-exposure bound over the 109 random one-dimensional targets of Table 3. The population is bimodal: most instances are tight to solver precision (gap $\lesssim 10^{-5}$), while the instances with gaps of 10^{-2}–10^{-1} have optimal-face rank 2 or 3 (Theorem 3).

Table 4: Rigorous separation of single- and multi-exposure optima by the interval branch-and-bound of Theorem 9 (DUV, coherent illumination). F: collected orders; SDP bound: rigorized certificate value t_R (optimum over any number of exposures); B&B L: certified lower bound on the single-mask optimum $\rho^\star(\mathcal{M}_\mathbb{R})$; best single U: incumbent, so that $L \leq \rho^\star(\mathcal{M}_\mathbb{R}) \leq U$; nodes: branch-and-bound nodes explored; converged: whether $L \geq (1 - \text{tol})U$ was reached; certified gain = $(L - \text{SDP bound})/\text{SDP bound}$, a rigorous lower bound on the multi-exposure improvement in ρ. "random n" are instances of the random one-dimensional set.

Case	F	SDP bound	B&B L	Best single U	Nodes	Converged	Certified gain
random 82 (coherent)	5	0.3597	0.4061	0.4066	40,383	yes	12.9%
random 91 (coherent)	5	0.6049	0.6775	0.6782	76,027	yes	12.0%
random 110 (coherent)	5	0.3464	0.4033	0.4037	17,413	yes	16.4%
random 50 (coherent)	5	0.3141	0.3343	0.3347	28,517	yes	6.4%
mixed (coherent)	7	0.4211	0.4776	0.4787	3.0×10^7	no (0.23% gap)	13.4%

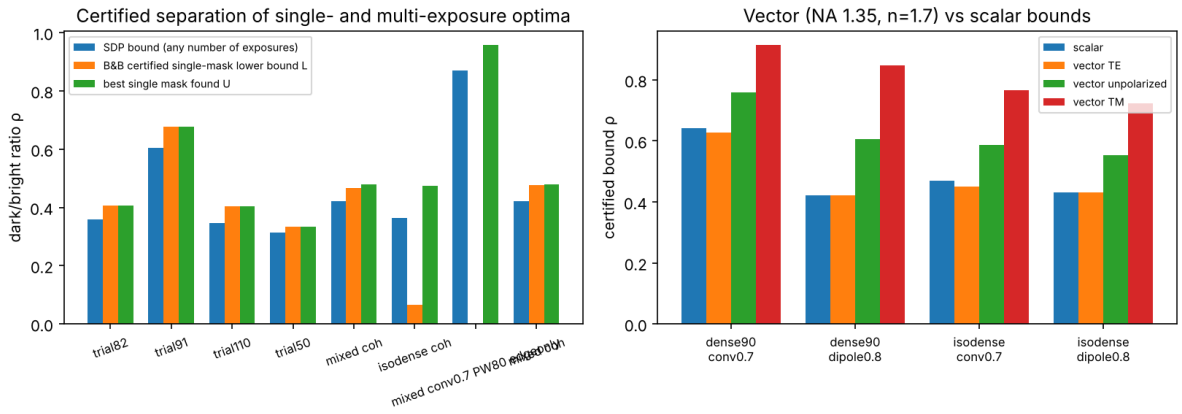


Figure 4: Left: certified separation of single- and multi-exposure optima (Table 4). For each case the bars show the SDP bound t_R (valid for any number of exposures), the branch-and-bound certified single-mask lower bound L, and the best single mask found U; a visible gap between the first bar and the second proves that multiple exposure strictly improves the ratio. For the two large-F cases (iso-dense coherent; mixed, $\sigma = 0.7$, ± 80 nm) the branch-and-bound did not converge and L is not informative. Right: certified bounds ρ under the scalar model and under the vector model (NA $= 1.35$, resist index $n = 1.7$) for TE, unpolarized and TM illumination on four one-dimensional targets (§6.9).

Table 5: Certified phase-shift (PSM) and binary-mask optima versus a standard pattern-error ILT (DUV, NA $= 1.35$, conventional $\sigma = 0.7$, $\tau = 10$ nm). All entries are dark-to-bright ratios ρ (lower is better). PSM bound: rigorized SDP certificate over real masks; binary bound: doubly-nonnegative (DNN) relaxation over nonnegative masks; binary penalty: certified relative increase of the binary bound over the PSM bound; ILT (PSM/binary): ratio reached by the pattern-error ILT mask; ILT refined: the ILT mask after local refinement on the ratio objective. "dense 120" is a 120 nm-pitch dense-line target.

Target	PSM bound	Binary bound	Binary penalty	ILT (PSM)	ILT (binary)	ILT refined
iso-dense	0.3879	0.4266	10%	0.553	0.579	0.426
mixed	0.4031	0.4599	14%	0.672	0.722	0.530
dense 120	0.3671	0.3777	3%	0.477	0.517	0.3671

6.6 Source–mask entitlement

For the iso-dense target the per-source-point relaxation equals the coherent bound (0.3625) both for $\sigma = 0.7$ and for a 13-point $\sigma = 0.9$ grid, whereas the fixed-source bounds are 0.43–0.51: no illumination supported on these points beats on-axis coherent light for this metric.

6.7 Two-dimensional layouts

Contact arrays are certified essentially tight with rank-one optima (Table 6). For the 320 nm cells ($F = 61$, approximately 1,200 constraints) the strong search closes the annular T-junction and the line-end gaps to 0.61% and 0.03% (Fig. 5); the quasar T-junction retains a 6.2% gap that we could neither close nor certify as a multi-exposure gain (the interval branch-and-bound is out of reach at $F = 61$, and a two-exposure local refinement did not beat the single mask). The weighted dark matrix has a four-dimensional common null space (orders invisible to every constraint); the certificate is verified on its complement. On 40 random two-rectangle layouts in 200–240 nm cells (annular, quasar and $\sigma = 0.5$ illumination; $F = 13$–37) the single-mask search matches the bound to 10^{-3} in 45% and to 5% in 62.5% of cases; the remaining gaps (up to 26%, mostly for $\sigma = 0.5$ and $F = 37$) are unresolved upper bounds on the multi-exposure benefit (Fig. 6, left). EUV contact arrays (32–44 nm pitch, ± 40 nm focus) are tight, with certified EL from 0.13 to 0.41 (Fig. 6, right).

Table 6: Two-dimensional layouts (DUV, $\lambda = 193\,\text{nm}$, NA $= 1.35$). Contact arrays: pitch in nm as given in the case name, with edge tolerance $\tau = 12$, 10 and 8 nm for 160, 120 and 100 nm pitch respectively; T-junction and line-end cells: 320 nm cell, $\tau = 14\,\text{nm}$. F: collected orders; bound ρ: rigorized certificate value; max EL: $(1-\rho)/(1+\rho)$ at the bound; best single ρ: best feasible real mask found; gap = (best single mask − bound)/bound; SDP time: wall-clock time of the relaxation solve in seconds. "strong": 12-start search with vectorized SLSQP polish.

Layout	F	Bound ρ	Max EL	Best single ρ	Gap	SDP time
contacts 160, annular	13	0.5843	0.262	0.5861	0.30%	1.4 s
contacts 160, $\sigma = 0.5$	9	0.6174	0.237	0.6177	0.04%	1.5 s
contacts 120, annular	9	0.6957	0.179	0.6962	0.07%	0.7 s
contacts 120, quasar	9	0.5740	0.271	0.5747	0.13%	0.7 s
contacts 100, quasar	5	0.8751	0.067	0.8754	0.03%	0.5 s
T-junction 320, annular (strong)	61	0.4387	0.390	0.4413	0.61%	410 s
T-junction 320, quasar (strong)	61	0.3893	0.440	0.4135	6.2%	269 s
line-ends 320, annular (strong)	61	0.5194	0.316	0.5195	0.03%	649 s

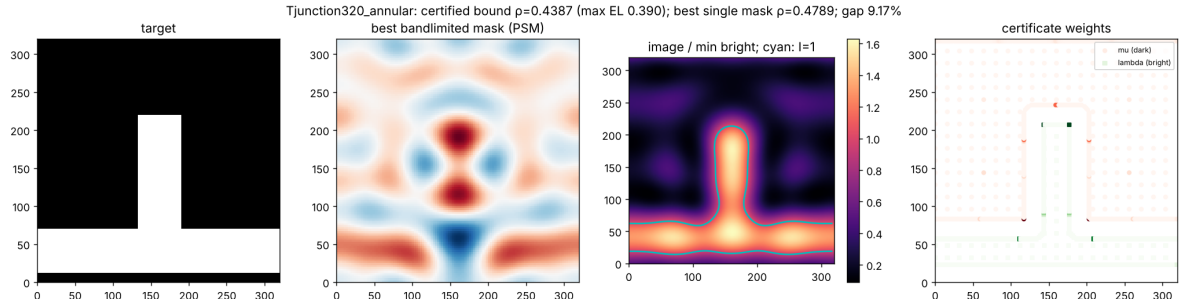


Figure 5: T-junction in a 320 nm cell under annular illumination (DUV, NA $= 1.35$, $\tau = 14\,\text{nm}$; axes in nm). From left to right: the target (white = bright); the best band-limited phase-shift mask (signed amplitude); the normalized image with the $I = 1$ contour (cyan), which traces the printed feature; and the certificate weights μ_d (dark) and λ_b (bright), which localize the limiting corners and the line-end. The certified bound is $\rho = 0.4387$ (maximum EL 0.390). The mask shown is from the baseline search ($\rho = 0.4789$, as in the panel title); the strong search of Table 6 reaches $\rho = 0.4413$ (gap 0.61%) against the same certificate.

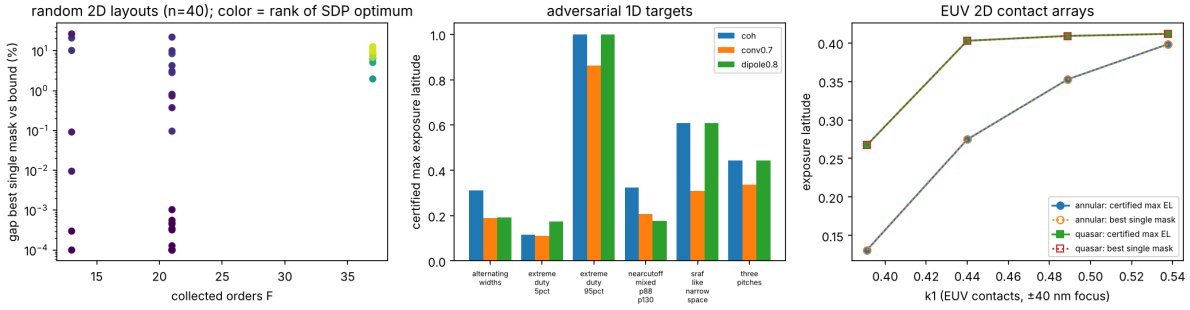


Figure 6: Expanded dataset. Left: gap between the best single mask and the certified bound (percent, logarithmic scale) versus the number of collected orders F for 40 random two-rectangle two-dimensional layouts; the marker color encodes the rank of the SDP optimum (dark = rank one). Middle: certified maximum exposure latitude of the adversarial one-dimensional targets of §6.8 under coherent ("coh"), conventional $\sigma = 0.7$ ("conv0.7") and dipole $\sigma = 0.8$ ("dipole0.8") illumination. Right: EUV two-dimensional contact arrays (32–44 nm pitch, ±40 nm focus) under annular and quasar illumination; the certified maximum EL and the best single mask coincide at every k_1.

Table 7: Sensitivity of the certified bound and solver validation (DUV, NA = 1.35; one-dimensional targets of Table 2 unless stated). Percentages are relative changes of the bound ρ with respect to the reference setting in the second column; EL is the exposure latitude $(1-\rho)/(1+\rho)$; κ is the rigorization scale of §5.

Study	Setting varied	Effect on the bound
interior sample spacing	16 to 1 nm	invariant (0.46907 throughout)
edge tolerance τ	2 to 20 nm	monotone: max EL 0.09 to 0.74
source quadrature	11 to 21 source points	converges: 0.513 to 0.526
complex- versus real-mask relaxation	nine cases	coincide (Lemma 6; Table 8)
Rayleigh-quotient bound	six cases	1–148× weaker (Table 8)
vector model, TE ($n_{\text{resist}} = 1.7$)	versus scalar	0–4% lower; tightness unchanged
vector model, unpolarized	versus scalar	10–40% higher
vector model, TM	versus scalar	70–120% higher (TM contrast loss)
pupil aberrations	0.02–0.1λ	iso-dense bound moves by 0–7%
pixelation	$N \geq F$ pixels per period	unchanged (Lemma 8)
pixelation	50 / 67 nm pixels	+15% / +46%
50-digit Cholesky of certificate	136 bounds	135 factorize; one has $S = 0$ (bound 1)
rigorization scale κ	136 bounds	> 0.99999, except one instance with $\kappa = 0.835$ (bound 10^{-3})
CLARABEL versus SCS	all instances	median difference 7×10^{-9}; five near-degenerate instances (bounds $\leq 10^{-2}$) off by up to 0.14, once negative

6.8 Adversarial one-dimensional targets

Near-cutoff mixed pitches, SRAF-like 20 nm spaces, 5%/95% duty cycles, alternating widths and three-pitch mixtures (Fig. 6, middle) are tight under dipole illumination, mostly tight under $\sigma = 0.7$, and show 12–37% gaps under coherent illumination (rank 3–5), the regime in which multiple exposure helps most. The isolated 20 nm space under coherent or dipole light has a certified bound of ≈ 0: a phase-edge mask prints an arbitrarily dark narrow line, which the relaxation recognizes.

6.9 Robustness, solver validation and physical sensitivity

Table 7 summarizes the sensitivity studies. The bound is invariant to the interior sample spacing and monotone in the edge tolerance τ; the source quadrature converges; the Rayleigh-quotient bound is 1–148× weaker (Table 8); and the complex- and real-mask relaxations coincide on all nine cases (Lemma 6). All 136 reported bounds were recomputed and their projected certificate matrices factorized exactly by a 50-digit Cholesky (135 succeed; the remaining EUV case, with bound exactly 1, has $S = 0$, Proposition 5); the rigorization scale κ exceeds 0.99999 except on one near-degenerate random instance (bound 10^{-3}, $\kappa = 0.835$). CLARABEL and SCS agree to a median of 7×10^{-9}; on five near-degenerate random instances (bounds $\leq 10^{-2}$) SCS is off by up to 0.14 and once negative, which the certificate check exposes. Under the vector model ($n_{\text{resist}} = 1.7$) the TE bounds are 0–4% below the scalar ones, the unpolarized bounds 10–40% above and the TM bounds 70–120% above (the classical TM contrast loss at NA = 1.35), with tightness unchanged (Fig. 4, right); aberrations of 0.02–0.1λ move the iso-dense bound by 0–7%; pixelation at $N \geq F$ leaves it unchanged and coarser pixels raise it (+15% at 50 nm, +46% at 67 nm), as Lemma 8 predicts.

Table 8: Certified SDP bound (real masks) versus the generalized-eigenvalue (Rayleigh-quotient) bound and the complex-mask relaxation on one-dimensional targets (DUV, NA = 1.35, $\tau = 8\,\mathrm{nm}$; dense 90: 6 nm). All entries are lower bounds on the dark-to-bright ratio ρ (higher is tighter); F: collected orders. The complex-mask relaxation coincides with the real-mask one in every case (Lemma 6).

Case	Source	F	SDP (real)	Rayleigh	SDP (complex)
iso-dense	coherent	9	0.3625	0.0668	0.3625
iso-dense	$\sigma = 0.7$	15	0.4691	0.1387	0.4691
iso-dense	dipole	15	0.4313	0.0823	0.4313
mixed	coherent	7	0.4211	0.0028	0.4211
mixed	$\sigma = 0.7$	11	0.4825	0.0343	0.4825
dense 90	$\sigma = 0.7$	9	0.6428	0.4392	0.6428

7 Discussion

The certificates turn folklore into checkable statements. Local optimization of the exposure-latitude objective is globally optimal for most one-dimensional targets and for the two-dimensional contact arrays; when it is not, the shortfall is—by Theorem 3—the benefit of incoherent multiple exposure, which Theorem 9 proves to be strictly positive on concrete coherent-illumination targets. Coherent illumination with phase-shift masks is the certified entitlement for the iso-dense target, and binary masks forfeit a certified 3–14%. Pattern-error ILT is far from EL-optimal, which argues for EL or contrast objectives, or for the certificate as a stopping criterion. The dual weights localize the limiting geometry. These conclusions survive the vector model and moderate aberrations. Finally, the relaxation gap has a physical meaning and is therefore informative rather than merely a weakness.

8 Limitations

The model is a thin-mask periodic model (no three-dimensional mask electromagnetics, resist chemistry, flare or mask error) with sampled constraints and an explicit edge tolerance. The objective is the ratio/EL objective (any constraint linear in intensity lifts identically; edge-placement error enters only through τ). The DNN relaxation is only an outer approximation in two dimensions; Lasserre-2 values carry a solver status and are not counted as certificates; the branch-and-bound converges only for $F \leq 7$; one 320 nm two-dimensional gap (6.2%) and the large-F random two-dimensional gaps remain unresolved; and near-degenerate instances (bound $\lesssim 10^{-2}$) stress first-order solvers. Literature verification relied on abstracts and search snippets (the full texts of [1, 7, 9] were not accessible).

Data and Code Availability

All code, data and the complete verification record accompany this manuscript in the repository directories `code/`, `results/` and `notes/`. The pipeline depends only on open-source Python packages (`pip install numpy scipy cvxpy matplotlib mpmath`). Running `cd code && python3 -W ignore test_certlitho.py` executes the nine unit tests, and `./run_all_par.sh` regenerates every table and figure of this manuscript from scratch (about 3.5 hours on four cores). The JSON outputs, solver logs and the table source `results/RESULTS_TABLES.md` are stored in `results/`.

Acknowledgements and Competing Interests

This manuscript was prepared by an autonomous research system under human oversight; the human overseers reviewed the claims, the verification record and the final text. No external funding was received for this work. The authors declare no competing interests.

References

[1] Alan E. Rosenbluth, David Melville, Kehan Tian, Kafai Lai, Nakgeuon Seong, Dirk Pfeiffer, and Matthew Colburn. Global optimization of masks, including film stack design to restore TM contrast in high NA TCC's. In *Proc. SPIE 6520, Optical Microlithography XX*, page 65200P, 2007.

[2] Anonymous. Pushing the limits of inverse lithography with generative reinforcement learning. arXiv:2602.19027, 2026.

[3] Linyong Pang. Inverse lithography technology: 30 years from concept to practical, full-chip reality. *Journal of Micro/Nanopatterning, Materials, and Metrology*, 20(3):030901, 2021.

[4] Hao Yang et al. Advancements and challenges in inverse lithography technology: a review of artificial intelligence-based approaches. *Light: Science & Applications*, 14:250, 2025. PMC12287475.

[5] Zhi-Quan Luo, Wing-Kin Ma, Anthony Man-Cho So, Yinyu Ye, and Shuzhong Zhang. Semidefinite relaxation of quadratic optimization problems. *IEEE Signal Processing Magazine*, 27(3):20–34, 2010.

[6] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

[7] Alan E. Rosenbluth, Scott J. Bukofsky, Carlos A. Fonseca, Michael S. Hibbs, Kafai Lai, Antoinette F. Molless, Rama Nand Singh, and Alfred K. K. Wong. Optimum mask and source patterns to print a given shape. *Journal of Microlithography, Microfabrication, and Microsystems*, 1(1):13–30, 2002. doi: 10.1117/1.1448500.

[8] Alan E. Rosenbluth et al. US patents 7,587,702 and 7,685,559 (step-walk relaxation method for global optimization of masks); 8,495,528 (optimized wavefronts for multiple exposure); 8,108,802 (arbitrary lithographic wavefronts); 7,944,545 (high contrast lithographic masks). United States Patent and Trademark Office, 2009–2014.

[9] Yong Liu and Avideh Zakhor. Binary and phase shifting mask design for optical lithography. *IEEE Transactions on Semiconductor Manufacturing*, 5(2):138–152, 1992. Bibliographic data partially confirmed (UC Berkeley EECS TR 1992).

[10] Nicholas D. Sidiropoulos, Timothy N. Davidson, and Zhi-Quan Luo. Transmit beamforming for physical-layer multicasting. *IEEE Transactions on Signal Processing*, 54(6):2239–2251, 2006.

[11] Gábor Pataki. On the rank of extreme matrices in semidefinite programs and the multiplicity of optimal eigenvalues. *Mathematics of Operations Research*, 23(2):339–358, 1998.

[12] Jean B. Lasserre. Global optimization with polynomials and the problem of moments. *SIAM Journal on Optimization*, 11(3):796–817, 2001.

[13] Guillermo Angeris, Jelena Vučković, and Stephen Boyd. Computational bounds for photonic design. *ACS Photonics*, 6(5):1232–1239, 2019.

[14] Guillermo Angeris, Jelena Vučković, and Stephen Boyd. Heuristic methods and performance bounds for photonic design. *Optics Express*, 29:2827–2854, 2021.

[15] Yukihide Kohira et al. Yield-aware mask assignment by positive semidefinite relaxation in triple patterning using cut process. *Journal of Micro/Nanolithography, MEMS, and MOEMS*, 15(2):021207, 2016.

[16] Xu Ma, Hao Zhang, Zhiqiang Wang, Yanqiu Li, Gonzalo R. Arce, Javier Garcia-Frias, and Lu Zhang. Information theoretical aspects in coherent optical lithography systems. *Optics Express*, 25(23):29043–29057, 2017.

[17] Zhiqiang Wang, Xu Ma, Gonzalo R. Arce, and Javier Garcia-Frias. Information theoretical approaches in computational lithography. *Optics Express*, 26(13):16736–16751, 2018.

[18] Emmanuel J. Candès, Thomas Strohmer, and Vladislav Voroninski. PhaseLift: exact and stable signal recovery from magnitude measurements via convex programming. *Communications on Pure and Applied Mathematics*, 66(8):1241–1274, 2013.

[19] Y. C. Pati and T. Kailath. Phase-shifting masks for microlithography: automated design and mask requirements. *Journal of the Optical Society of America A*, 11(9):2438–2452, 1994.

[20] Ulf Grenander and Gábor Szegő. *Toeplitz Forms and Their Applications*. University of California Press, 1958.

[21] Donis G. Flagello, Tom Milster, and Alan E. Rosenbluth. Theory of high-NA imaging in homogeneous thin films. *Journal of the Optical Society of America A*, 13(1):53–64, 1996.

[22] Paul J. Goulart and Yuwen Chen. Clarabel: an interior-point solver for conic programs with quadratic objectives. arXiv:2405.12762, 2024.

[23] Brendan O'Donoghue, Eric Chu, Neal Parikh, and Stephen Boyd. Conic optimization via operator splitting and homogeneous self-dual embedding. *Journal of Optimization Theory and Applications*, 169(3):1042–1068, 2016.